

OSTEOPOROSIS

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SCREENING AND DIAGNOSIS

Who to Screen?

- All post-menopausal females with OSTA (Age minus weight) > 20
- Post-menopausal females and males ≥ 65 years old with risk factors:
 - Non-modifiable: family history of osteoporosis or fragility fractures, history of falls, prolonged immobility, height loss (>2cm within 3 years), early menopause (45 years old and younger), presence of diseases that can lower bone density or increase fracture risk (e.g. prolonged untreated hyperthyroidism, inflammatory rheumatic disease, diabetes mellitus, etc.)
 - Modifiable: low elemental calcium intake (i.e. 800mg/day for adults 50 years and younger, <1000mg/day for adults older than 50 years old), excessive alcohol intake (>2 units per day for men, >1 unit per day for women), smoking (any), low BMI, certain medications (e.g. 5mg or more of prednisolone/day, androgen deprivation therapy, tamoxifen in premenopausal women, proton pump inhibitors)

Gold Standard Test for Diagnosis

Dual-Energy X-Ray Absorptiometry (DXA) of the hip and lumbar spine

Diagnosis of Osteoporosis

In male patients ≥ 50 years old or postmenopausal female patients:

1. BMD T-score ≤ -2.5 SD in femoral neck, total hip, or lumbar spine and/or
2. history of fragility fracture

When to Consider Treatment

1. Osteoporosis diagnosed based on BMD T-score.
2. For fragility hip or vertebral fractures, regardless of BMD results.
3. For BMD T-score between -1.0 and -2.5, calculate FRAX score and use the following thresholds:

 Consider initiating treatment when the probability of fracture exceeds...	Patient with BMD T-score between -1.0 and -2.5:			
	Female		Male	
	 50–69 years old	 70+ years old	 50–69 years old	 70+ years old
10-year hip fracture probability threshold	≥3%	≥4%	≥3%	≥4%
OR	OR	OR	OR	OR
10-year major osteoporotic fracture probability threshold	≥18%	≥18%	≥15%	≥15%

For patients with BMD T-score between -1.0 and -2.5 who are not initiated on treatment, optimise lifestyle management and periodically reassess fracture risk.

For fragility fractures at other sites (excluding fingers and toes), please use clinical judgement, BMD and FRAX scores for guidance.

N.B.: *Fragility fracture is defined as a fracture that occurs as a result of minimal trauma, such as a fall from a standing height or less, or no identifiable trauma.*

INITIAL ASSESSMENT AND MANAGEMENT

General Management

1. Educate patient about osteoporosis and fragility fractures and their implications.
2. Refer to Care Manager for assessment and education regarding:
 - Fall risk, home safety and footwear.
 - Appropriate weight bearing and muscle strengthening exercises (e.g., walking, elastic band exercises).
 - Calcium / Vitamin D diet advice.
 - A 2nd visit is recommended to review fall risk and diet history.
3. Advise smoking cessation, reduce alcohol intake.
4. Calcium and Vitamin D supplements if not meeting recommended dietary allowance.
5. Pharmacotherapy where indicated.

FRAX® Tool

The FRAX® tool has been developed by WHO to evaluate fracture risk of patients. It is based on individual patient models that integrate the risks associated with clinical risk factors as well as bone mineral density (BMD) at the femoral neck (in g/cm²).

Notes for use of FRAX:

- *Select the correct country and race.*
 - *Select “yes” for glucocorticoids if patient has at least 5mg of daily prednisolone for at least 3 months.*
 - *Select “yes” for rheumatoid arthritis if patient has Diabetes mellitus (DM is associated with increase in fracture risk).*
 - *For femoral neck BMD, select “Hologic” for the DXA equipment and enter the actual femoral neck BMD (in g/cm²) instead of T-score.*
- FRAX score may also be calculated without a BMD (to leave the field blank).*

Available at www.shef.ac.uk/FRAX

Can be found in intranet under Doctors → Clinical Guidelines & Protocol → Endocrine: [Charts & Calculators](#)

Pharmacological Treatment

Consider oral alendronate or risedronate for treatment. SC denosumab and IV zoledronate can be considered when oral bisphosphonates are not suitable or not preferred.

For patients at very high risk of fractures (BMD with T-score ≤ -3.0) AND ≥ 2 vertebral fractures or ≥ 2 fragility fractures), consider referral for anabolic agents (e.g. teriparatide or romosozumab) as first-line treatment.

Before initiating treatment, consider:

1. Secondary causes for osteoporosis: Hypogonadism, Hyperthyroidism, Hyperparathyroidism, Cushing's syndrome, Chronic Liver Disease, Chronic Renal Failure, Malabsorption, Rheumatoid Arthritis, Chronic Obstructive Lung disease and Anorexia Nervosa.
2. Screen baseline labs: FBC, Creatinine, Calcium, Phosphate, LFT, thyroid panel, 25(OH)D.
3. Assess the dental condition of the patient and avoid delaying osteoporosis treatment in patients with a fragility fracture or severe osteoporosis.
 - Inform of risks of medication-related osteonecrosis of the jaw (MRONJ) and atypical fractures with treatment but counsel that these are extremely rare.
 - Dental clearance is not routinely necessary in patients with good dentition.
 - Educate about the importance of good oral hygiene practices and arrange for regular dental reviews during treatment instead.
 - Offer dental clearance for the following group of patients:
 - Patients at higher risk of MRONJ (e.g. patients with cancer undergoing treatment, immunosuppression, choosing parenteral osteoporosis treatment)
 - Patients with concomitant periodontal disease or poor dentition that may require invasive dental procedures (e.g. extractions, implants)
4. Ensure adequate serum levels of Vitamin D (≥ 30 ng/mL) and Calcium. Consider starting calcium and vit D if necessary. **Treatment can be initiated concurrently with vitamin D replacement for levels ≥ 20 ng/mL.**
5. Ensure no contraindications to pharmacological agents (see below).

Drug	Evidence of effectiveness in reducing risk of: ^{15,28–31}	Contraindications ¹	Clinical considerations/precautions ¹
Alendronate, risedronate or ibandronate ⁱ (bisphosphonates, oral)	✓ Vertebral fracture ✓ Non-vertebral fracture (alendronate & risedronate only) ✓ Hip fracture (alendronate & risedronate only)	• CrCl <35 mL/minute (alendronate) or <30 mL/minute (risedronate, ibandronate) • Hypocalcaemia • Oesophageal abnormalities such as strictures, achalasia, or uncontrolled gastro-oesophageal reflux disease • Inability to stand or sit upright for at least 30 minutes	Advise patients to: • Swallow the whole tablet early in the morning on an empty stomach with plain water, at least 30 minutes before (for alendronate and risedronate) or at least 60 minutes before (for ibandronate) consuming any food, beverages, or other medications (especially antacids, calcium, iron, or mineral supplements) • Remain upright (can sit, stand or walk) and avoid lying down for at least 30 minutes before eating the first meal of the day
Zoledronateⁱ (bisphosphonate, IV)	✓ Vertebral fracture ✓ Non-vertebral fracture ✓ Hip fracture	• CrCl <35 mL/minute • Hypocalcaemia	• Give IV infusion for at least 15 minutes • Ensure patient is adequately hydrated before use • Use with caution in patients with significant vitamin D deficiency. Consider completing loading dose of vitamin D prior to starting zoledronate • Check serum calcium and phosphate at 9 to 14 days after infusion, if patient shows symptoms of hypocalcaemia or hypophosphataemia
Denosumabⁱ (RANKL inhibitor, SC)	✓ Vertebral fracture ✓ Non-vertebral fracture ✓ Hip fracture	• Hypocalcaemia	• Compliance to 6-monthly SC injection is important • For patients with vitamin D deficiency, consider completing loading dose of vitamin D prior to starting denosumab (see Table 2) • Exercise caution in patients with preexisting eczema and recurrent infections (e.g. cellulitis)

MONITORING AND LONG TERM FOLLOW UP

Monitoring

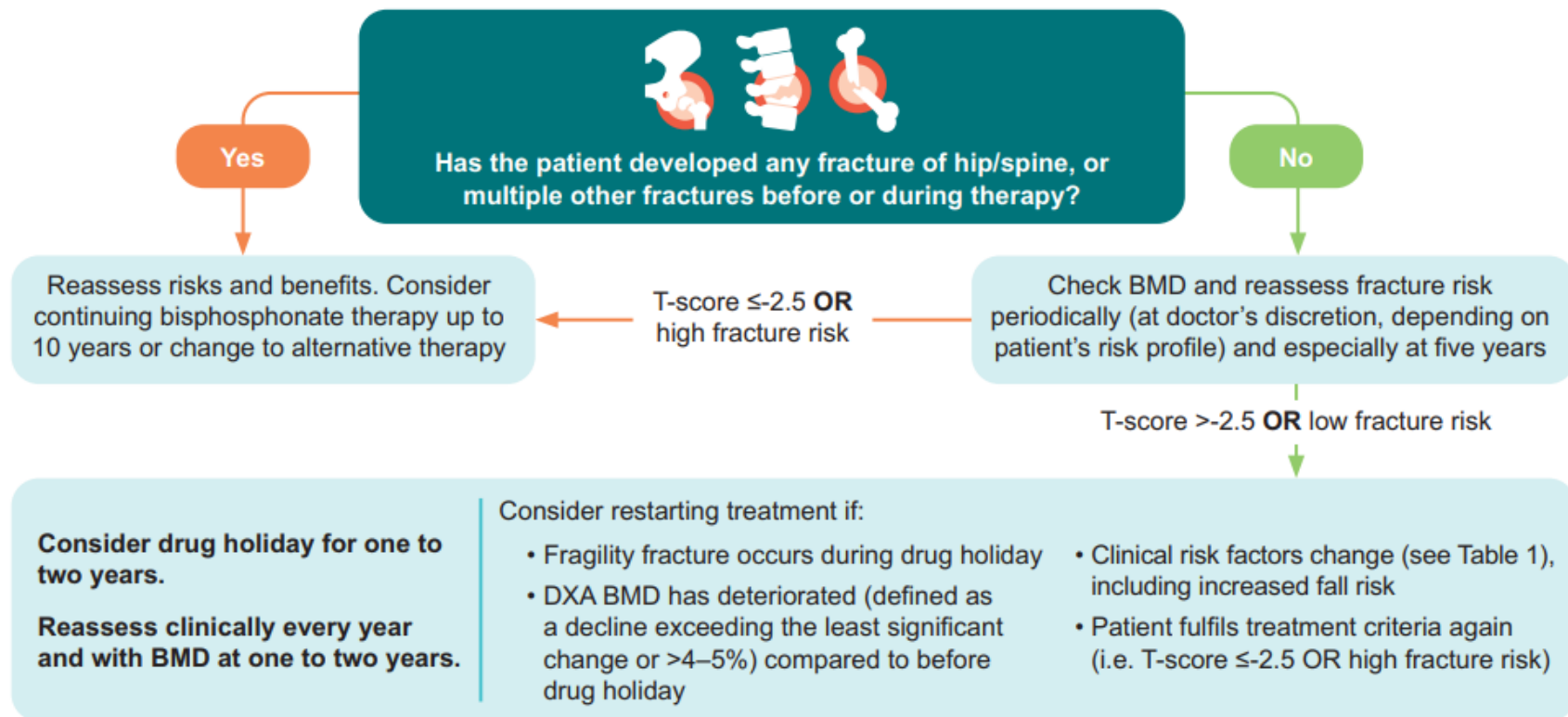
1. DXA-BMD should be done at baseline. However, in patients with a history of fragility fractures, pharmacological treatment should not be delayed while awaiting BMD results. Repeat BMD should be considered 2 years after initiation of treatment to evaluate clinical effectiveness. Subsequent BMD must be performed every 2 years to assess for deterioration (decline exceeding least significant change or >4-5%).
2. Assess Creatinine and Calcium levels annually.
3. Assess for any fracture occurring whilst on medication.
4. Assess for adverse effects associated with prolonged treatment with bisphosphonates:
 - Non-specific gastrointestinal symptoms (e.g., nausea, dyspepsia, abdominal pain, reflux)
 - Trial bisphosphonate with PPI (omeprazole 20 mg) taken the night before
 - Osteonecrosis of jaw (rare, but important side effect, encourage routine dental visits)
 - Atypical femoral fractures (to be vigilant especially if patient complains of thigh pain)
5. Dental procedures while on treatment:
 - Temporary discontinuation of bisphosphonates before invasive dental procedures is generally not required but may be considered for patients at higher risk of MRONJ.
 - For patients on denosumab, invasive dental procedures may be timed at least 2 – 4 weeks before the next scheduled 6-monthly injection to support mucosal healing.
 - Clinical decisions should prioritise individual risk assessment and preventive care, as there is no strong evidence that stopping treatment reduces MRONJ risk – and doing so must be balanced against the risk of fractures.

Drug Holiday Depends on Risk Stratification for Recurrent Fractures

1. Antiresorptive effects of bisphosphonates persist beyond discontinuation and a drug holiday may be considered for patients after five years of bisphosphonate therapy.
2. Duration of treatment and duration of drug holiday depends on individual patient's fracture risk, T-score, and history of fracture.

Who would be suitable: Patients with a repeat T-score above -2.5 SD and no new fractures at 5-year post therapy.

N.B.: A reasonable approach for patients with low risk of fracture is to stop treatment after five years and remain off as long as bone mineral density is stable and patient has no previous vertebral fractures.



Drug holidays are not recommended for patients on denosumab, due to risk of bone loss upon cessation. If denosumab needs to be stopped or delayed, a plan to transition to a bisphosphonate should be in place, where appropriate.

CALCIUM AND VITAMIN D

Diet and Supplements

It is important to assess patient's dietary intake of calcium / Vitamin D before recommending appropriate calcium / Vitamin D supplements.

Food	Serving Size	Calcium content (mg)
Dairy Products		
High-calcium milk powder	4 scoops (25 g)	500
Low-fat milk	1 glass (250 ml)	380
Full-cream milk	1 glass (250 ml)	300
Low-fat yoghurt	1 carton (150 g)	240
Low-fat cheese*	1 slice (20 g)	200
Non-Dairy Foods		
Canned sardine (with bones)	1 fish (80 g)	270
Dried ikan bilis (with bones)	2 tablespoons (40 g)	270
Silken tofu	package (150 g)	100
Tau kwa	1 small cake (90 g)	150
Dhal (raw)	mug (50 g)	85
Baked beans, canned*	can (210 g)	110
Kai lan, cooked	mug (100 g)	195
Spinach, cooked	mug (100 g)	140
Chye sim, cooked	mug (100 g)	140
Broccoli, cooked	mug (100 g)	50
Calcium-Fortified Products		
High-calcium soybean milk	1 glass (250 ml)	450
Enriched bread	2 slices (60 g)	100
Calcium-fortified orange juice	1 serving (240 ml)	350

* Item may be high in salt

Dietary sources of Vitamin D: Vitamin D fortified cereal / milk / orange juice, cod liver oil, egg yolk, salmon, tuna, sardines

MEDICATIONS

Common Calcium Formulations

Formulations	Suggested Dose	Common ADR	Contraindications / Precautions / Remarks
Calcium Carbonate (S1) Strength per tablet (<i>Elemental Ca</i>): 625 mg (250 mg), 1,250 mg (500 mg) \$^A	625 - 1250 mg OM	Can cause constipation and flatulence	Should be taken with meals as acidity improves absorption. Commonly used calcium supplement. Drug Interactions: Space calcium 2 hours apart from medications such as iron supplements, levothyroxine, and certain antibiotics (quinolones, tetracyclines).
Calcium Carbonate / Vitamin D3 (S1) Strength per tablet (<i>Elemental Ca</i>): CaCO ₃ 450 mg (180 mg); Vit. D3 200 IU \$^A	1 - 2 tabs OM		Take with or after food. May be used with long-term gastric acid suppression agents.
Calcium Acetate (S1) Strength per tablet (<i>Elemental Ca</i>): 667 mg (169 mg) \$-\$\$\$^A	667 mg BD-TDS (with meals)		Mainly use in the treatment of hyperphosphatemia in end stage renal disease

N.B.: Most OTC supplements' calcium content is addressed in elemental calcium level. **The recommended dietary intake of elemental calcium for healthy adults 51 years and older is 1,000 mg per day.** (For patients with osteoporosis, recommended intake is 1,200 mg per day for female age > 50 and male age > 70)

^ Cost \$ <\$10 per month \$\$ \$10 - <\$20 per month \$\$\$ \$20 - <\$30 per month
 \$\$\$\$ \$30 - <\$60 per month \$\$\$\$\$ \$60 - <\$90 per month \$\$\$\$\$\$ \$90 or more per month

Amount payable depends on patient subsidy level and drug subsidy class (Standard Drug List S1 and S2, Non-Standard Drug NS)

Vitamin D Formulation

Formulations	Suggested Dose	Common ADR	Contraindications / Precautions / Remarks
Vitamin D3 = Cholecalciferol (S1) Strength per tablet: 1,000 IU \$^	Maintenance Dose 1,000 IU/day		
D-Cure = liquid formulation of vitamin D3 cholecalciferol (NS) Oral liquid formulation: 25,000 IU per 1 ml ampoule \$\$-\$\$\$^	Initial Replacement Dose 50,000 IU/week for 6-8 weeks for initial replacement of vitamin D deficiency Maintenance Dose 25,000 IU/month		Take with food. Better absorption compared to D2. Caution re: Vitamin D overdose, i.e., remind patients to switch to monthly dose for maintenance therapy after 8 weeks instead of continuing weekly dose.
Vitamin D2 = ergocalciferol Strength per capsule: 50,000 IU (Available OTC)	Initial Replacement Dose 50,000 IU/week (during initial replacement for Vitamin D deficiency)		Available in few OTC vegan Calcium / Vitamin D preparations
Alfacalcidol = 1hydroxycholecalciferol (S2) Strength per tablet: 0.25 mcg \$\$\$^	Maintenance Dose 0.25 mcg – 1 mcg/day		Be cautious of hypercalcaemia
Calcitriol = 1,25- dihydroxyvitamin D (S2) Strength per capsule: 0.25 mcg \$\$\$^	Maintenance Dose 0.25 mcg/day		Be cautious of hypercalcemia and hypervitaminosis D. Drug of choice in advanced or end stage renal disease patients. To be initiated and titrated according to PTH levels at specialist level.

^ Cost \$ <\$10 per month \$\$ \$10 - <\$20 per month \$\$\$ \$20 - <\$30 per month
 \$\$\$\$ \$30 - <\$60 per month \$\$\$\$\$ \$60 - <\$90 per month \$\$\$\$\$\$ \$90 or more per month

Amount payable depends on patient subsidy level and drug subsidy class (Standard Drug List S1 and S2, Non-Standard Drug NS)

Anti-Resorptive Treatments

Medications	Dose	Common ADR	Contraindications / Precautions / Remarks
Bisphosphonates: Alendronate 70mg tab (Fosamax®) (S2) \$^ Risedronate 35mg tab (Actonel®) (S2) \$^ Risedronate 150mg tab (Actonel®) Once a month dosing (Not available in NUP)	Alendronate: 70mg once weekly Risedronate: 35 mg once weekly	Reflux, esophagitis, ulcers Osteonecrosis of the jaw (ONJ) (rare, to encourage good oral hygiene and regular dental visits) Risk: Atypical femoral fractures. Hypocalcaemia	Indication: Reduce vertebral, non-vertebral and hip fractures. Counselling points: Swallow tablet whole with 1 glass of water at least 30 minutes before taking food, beverages, or other medications (esp. antacids, calcium, iron, or mineral supplements). Remain upright for at least 30 minutes after eating. Contraindications: (Alendronate: CrCl < 35 ml/min, Risedronate: CrCl < 30ml/min) Oesophageal abnormalities (Barrett's, stricture, achalasia). Hypocalcaemia (ensure adequate Calcium and Vit D intake). Unable to sit or stand for >30 minutes. Aspiration risk and difficulty swallowing liquids.
Denosumab 60mg/ml Inj (Prolia®) (NS) \$\$\$\$^# MAF eligible in patients with osteoporosis at high risk of fracture	SQ 60 mg once every 6 months	Arthralgia, back pain constipation, diarrhoea, nausea hypercholesterolemia, eczema, dermatitis, risk of infection (including skin infection, cystitis), headache, fatigue, anaemia, thrombocytopenia Risk: Atypical femoral fractures, ONJ and hypocalcaemia with renal impairment.	Indication: Reduce vertebral, non-vertebral and hip fractures. Reduces bone resorption, increase bone mass and strength. Can reduce cancer-related bone pain. Cost is high, usually initiated as 2 nd line by specialists. Contraindications: Hypersensitivity. Hypocalcaemia (ensure adequate Calcium and Vit D intake). Pregnancy.

^ Cost \$ <\$10 per month

\$ \$ \$10 - <\$20 per month

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\$ \$ \$ \$ \$ \$90 or more per month

Amount payable depends on patient subsidy level and drug subsidy class (Standard Drug List S1 and S2, Medication Assistance Fund, Non-Standard Drug NS)

Average monthly cost

CRITERIA FOR REFERRAL TO TERTIARY CARE

Endocrinology

1. Male with fragility fracture < 65 years old
2. Pre-menopausal with fragility fractures or Z-score < -2.0
3. Very high risk of fractures (multiple fragility fractures and T-score < -3.0 (may be considered for anabolic agents))
4. Poor response to bisphosphonates despite compliance for at least 1 year (new fractures or decreasing BMD trend of > 4-5%)
5. Unable to tolerate oral bisphosphonate therapy:
 - Renal impairment: ***If the calculated creatinine clearance (Cockcroft-Gault) is persistently less than 30ml/min or CrCl 30-60ml/min with abnormal calcium / phosphate levels, refer to renal specialist clinic***
 - Gastrointestinal side effects (gastritis / reflux / PUD)
 - Atypical fractures or h/o osteonecrosis of jaw whilst on bisphosphonates
6. Suspected complex secondary causes of osteoporosis due to Endocrine disorders:
 - E.g., Hyperparathyroidism, Cushing's syndrome, hypophosphatemia, male hypogonadism, prolactinoma, hypopituitarism
7. Women with pregnancy- and lactation-associated osteoporosis.

Other Specialties

1. Rheumatology (Connective tissue disease)
2. Haematology (Myeloma)
3. Gastroenterology (Malabsorption syndrome)
4. Falls Clinic / GRM / Neurology (recurrent falls)

Second Line Treatment (Usually Started by Specialist)

1. Oral Raloxifene (Selective Estrogen Receptor Modulator)
2. SC Denosumab once / 6-monthly (monoclonal antibody)
3. SC Romosozumab (monoclonal antibody: sclerostin inhibitor) / monthly for up to 1 year
4. IV Zoledronate once/ year (IV bisphosphonate)
5. SC Teriparatide daily (recombinant parathyroid hormone)

RECOMMENDED CARE COMPONENTS FOR OSTEOPOROSIS

Recommended Care Components	Minimum Frequency	Remarks
Dual-Energy X-Ray Absorptiometry (DXA) scan [^]	At least once every 2 years	For treatment monitoring, consider DXA at baseline, after two years of treatment (to establish clinical effectiveness), and every two years thereafter. Clinicians may review the frequency of DXA assessment accordingly based on patients' risk factors and response to treatment.
Serum creatinine and calcium	Annually while on pharmacotherapy	To monitor CrCl not contraindicating pharmacotherapy and for hypocalcaemia while on anti-resorptives.

[^] *These are non-reportable clinical indicators*

Who to Screen:

- Post-menopausal females and males ≥ 65 years old with risk factors
- Postmenopausal female with OSTA (Age minus weight) > 20
- Examples of risk factors:
 - * *prolonged steroid use (≥ 5 mg prednisolone/day or equivalent for > 3 months)*
 - * *early menopause (45 years old and younger)*
 - * *family history of osteoporosis*
 - * *prolonged immobilisation*

OSTEOPOROSIS MANAGEMENT IN PRIMARY CARE

Diagnosis of Osteoporosis:

Patients > 50 years old or postmenopausal patients with:
DXA-BMD T-score ≤ -2.5 SD in femoral neck, total hip or lumbar spine
or
History of fragility fractures (hip or spine with minimal trauma)

General Management:

1. Refer to Care Manager for:
 - Fall risk, home safety and footwear assessment
 - Teach weight bearing and muscle strengthening exercises
 - Calcium / Vitamin D dietary advice
 - Annual visit is recommended to review fall risk and diet history
2. Advise smoking cessation, reduce alcohol intake
3. Calcium and Vitamin D supplements as necessary

Criteria for Specialist Referral (Endocrinology):

1. Male with fragility fracture < 65 years old
2. Pre-menopausal with fractures or Z-score < -2.0 SD
3. Multiple fragility fractures with T-score ≤ -3.0 SD
4. Poor response to bisphosphonate therapy despite compliance: New fractures or Decrease BMD $> 4-5\%$
5. Contraindications to bisphosphonate therapy:
 - Renal impairment: CrCl (Cockcroft-Gault[^]) < 30 ml/min or CrCl 30-60 ml/min with abnormal calcium / phosphate levels (refer renal as well)
 - Gastrointestinal side effects (gastritis / reflux / PUD)
6. Suspected complex secondary causes of osteoporosis:
 - Endocrinology (Cushing's syndrome, hyperparathyroidism, hypophosphatemia, male hypogonadism)
7. Women with pregnancy- and lactation-associated osteoporosis.

Criteria for Referral to Other Specialists:

- Rheumatology (Connective tissue disease)
- Haematology (Myeloma)
- Gastroenterology (Malabsorption syndrome)
- Falls Clinic / GRM / Neurology (Recurrent falls)
- Renal (Renal impairment)

First Line Treatment:

Oral bisphosphonates (Alendronate or Risedronate)

- Before starting:
 - Consider secondary causes* for osteoporosis
 - Screen baseline: FBC, Creatinine calcium, phosphate, 25(OH)D, LFT, thyroid panel
 - Refer dental if poor dentition
 - Ensure vitamin D and calcium replete
- Monitoring:
 - Assess for side effects: heartburn, nausea, MSK aches, rare cases of osteonecrosis of jaw, atypical femoral fractures
 - Creatinine and calcium levels annually
 - BMD at baseline and 2 yearly later to assess response to therapy

Second line treatment: (to be started by specialist)

Oral Raloxifene
SC Denosumab once/6-monthly
SC Romosozumab monthly for up to 1year
SC Teriparatide daily
IV Zoledronate once/year

**Medical conditions that may cause secondary osteoporosis:*

Hyperthyroidism, Hypogonadism, Hyperparathyroidism, Cushing's Disease, Chronic Liver Disease, Chronic Renal Failure, Malabsorption, Rheumatoid Arthritis, Chronic Obstructive Lung disease and Anorexia Nervosa

When to Start Therapy:

- History of fragility fractures
- T-score below -2.5 SD
- 10-year fracture risk using FRAX[®] exceeding treatment thresholds
- Consider individual patient factors: patient's preference, comorbidities, risk factors not captured in FRAX model, side effects and cost of medications.

Drug Holiday:

It is recommended to consider a drug holiday after 5 years of bisphosphonate therapy.

Duration of drug holiday: 1 to 2 years

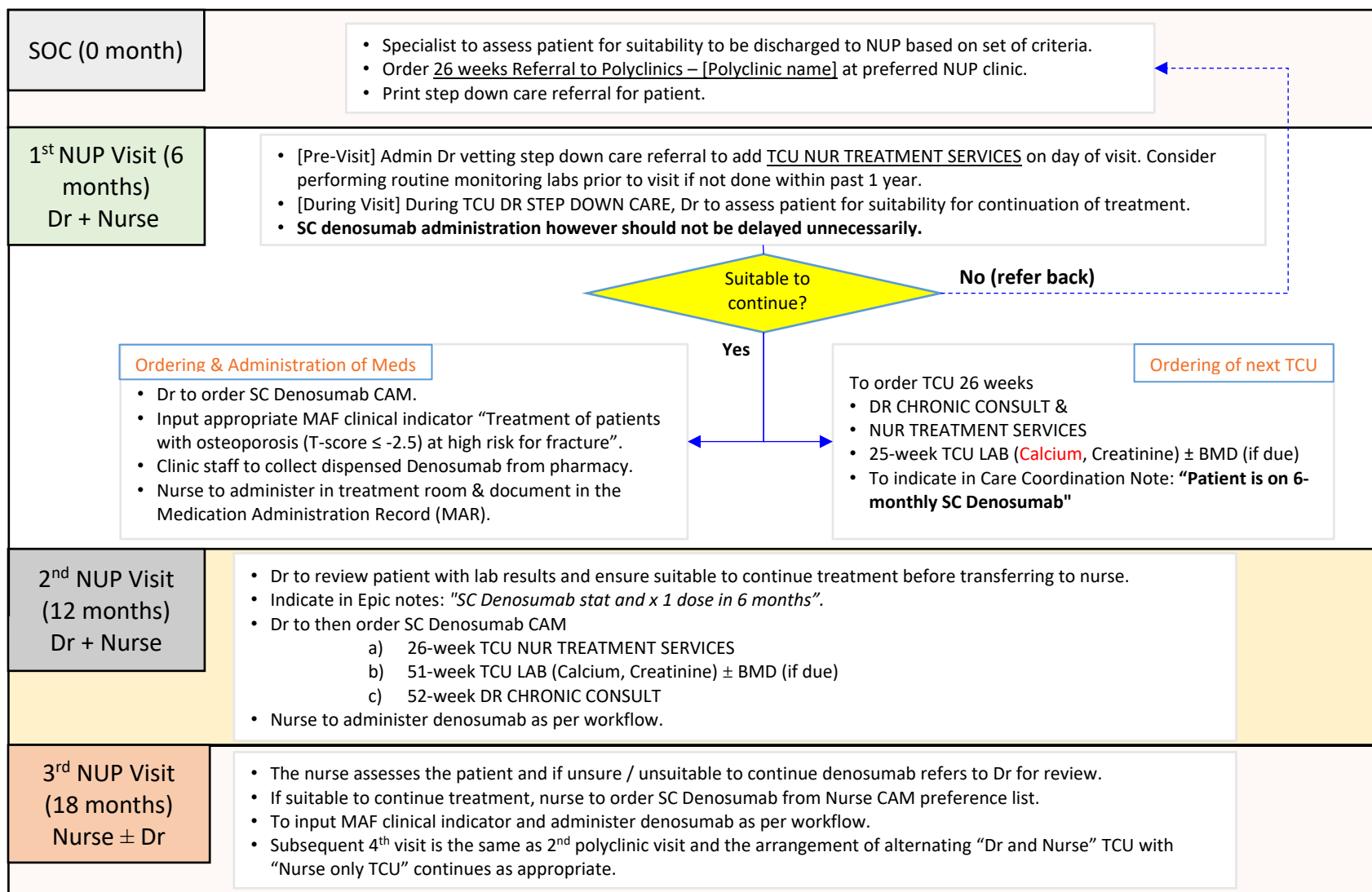
Who would be suitable: Patients with a repeat T-score above -2.5 SD and no new fractures at 5-year post therapy.

NB: can treat up to 10yrs for pts with high fracture risk.

[^] Use ideal body weight for patients who are overweight.

RIGHT SITING OF DENOSUMAB TO NUP

Since October 2022, NUP has established a workflow to allow SOC patients who have received at least **1** doses of Denosumab SC injection to be decanted to NUP. The pilot workflow will allow polyclinics to receive patients on Denosumab injections after the medication has been safely initiated by specialists.



NUP doctor will review patient annually to ensure suitability for continuation of Denosumab in polyclinic, and nurses will review 6-mthly to continue Denosumab SC injection.

1. **Doctor's Responsibility**

- a) To assess patient's referral criteria and any contraindications / side effects to Denosumab
 - i. Side effects: Back pain, musculoskeletal pain, constipation, cystitis, risk of hypocalcaemia in CKD3 and above, risk of serious skin infection 1:200, rare risk of atypical femoral fractures or osteonecrosis of jaw.
 - ii. Postpone injection by 2 weeks if patient has fever or active infection on day of injection.
- b) To ensure patient has sufficient calcium / vitamin D supplements. All patients should receive calcium 1,000 mg daily and at least 400 IU vitamin D daily whilst undergoing treatment.
- c) To ensure Calcium, Creatinine and Vitamin D are done within the last 1 year, admin doctor vetting step down referral to NUP to consider ordering these tests prior to 1st visit.
- d) To ensure date of last injection is at least 20 weeks from current. To advise patient not to delay the next injection by more than 32 weeks as there may be an increased risk of vertebral fractures with a delayed dose.
- e) Refer to specialist clinic if re-referral criteria are met.
- f) Order TCU for Subsequent visits:
 - i. 26-week TCU for DR CHRONIC CONSULT + NUR TREATMENT SERVICES + 25-week TCU Lab as last denosumab panel would have been done at SOC (Vitamin D, Calcium, Creatinine) and/or BMD (if due)
 - ii. Thereafter, reviewing doctor can order subsequent follow-up:
 - 26-week TCU for NUR TREATMENT SERVICES and
 - 52-week TCU for DR CHRONIC CONSULT + NUR TREATMENT SERVICES + 51-week TCU Lab (Vitamin D, Calcium, Creatinine) KIV add Vitamin D hydroxy if no improvement in BMD
 - **Yearly monitoring of Vitamin D, Calcium, Creatinine for patients on Denosumab. Consider 6 monthly monitoring if patient has CKD 3 or suspected non-adherence to calcium/vitamin D supplementation.**
- g) To refer to financial counsellor if patient needs financial assistance.

2. **Conversion of Denosumab to oral bisphosphonate in NUP may be considered if:**

- a) T-score has improved (> -2.5) and no new fractures
 - b) Patient has adverse effects to Denosumab
 - c) Do not stop Denosumab in polyclinics if patient is on aromatase inhibitors, androgen deprivation therapy, or high dose steroids. If patients wish to stop Denosumab whilst being prescribed these drugs, do refer patients back to the hospital.
- 2.1 Inform patient of the risk of rapid BMD decline if Denosumab is delayed or discontinued.
- 2.2 There is a need to switch to oral bisphosphonates or alternative therapy after stopping Denosumab. E.g., If a patient has T-score > -2.5 on denosumab and is keen for a drug holiday, first transition to an oral bisphosphonate for at least 2 years while ensuring the BMD is stable.
- 2.3 Check BMD 2 yearly and continue oral bisphosphonate as necessary.
- 2.4 If patient is unable to tolerate oral bisphosphonates, or has been on Denosumab for > 10 years, or had recent major fragility fracture (within 2 years), to refer to Specialist for alternative therapy.

3. **Criteria for Re-referral to respective Osteoporosis Specialist**

- ≥ 2 new fragility fractures after Denosumab treatment was initiated.
- ≥ 1 new fragility fracture after treatment was initiated AND declining BMD beyond Least Significant Change (i.e., $> 4\text{-}5\%$ decrease)
- Declining BMD beyond Least Significant Change (i.e., $> 4\text{-}5\%$ decrease) over 2 years despite adherence to Denosumab therapy
- Calculated creatinine clearance (by Cockcroft-Gault) persistently less than 30ml/min $\rightarrow$ to refer to Nephrology as well to exclude CKD-related metabolic bone disease.
- T-score has improved > -2.5 AND no new fragility fractures within 2 years (i.e., ready to stop Denosumab), but unable to tolerate oral bisphosphonates.
- T-score has improved > -2.5 AND no new fragility fractures within 2 years (i.e., ready to stop Denosumab), after more than 5 years of Denosumab use, but unable to tolerate oral bisphosphonates.
- T-score has improved > -2.5 , with recent major fragility fractures (i.e., hip, vertebral, pelvic, proximal humerus fracture) within past 2 years
- Use of Denosumab for more than 10 years and unable to tolerate oral bisphosphonates.
- Development of conditions associated with low bone mass (e.g., Cushing's syndrome, hypogonadism, and hyperparathyroidism)
- Initiation of medications associated with low bone mass (e.g., steroids, androgen deprivation therapy, aromatase inhibitors)

Nb. Patients with CKD 4 or 5 should be managed at SOC.

CALCIUM METABOLISM

HYPERCALCAEMIA

Symptoms and Causes

Physical manifestations

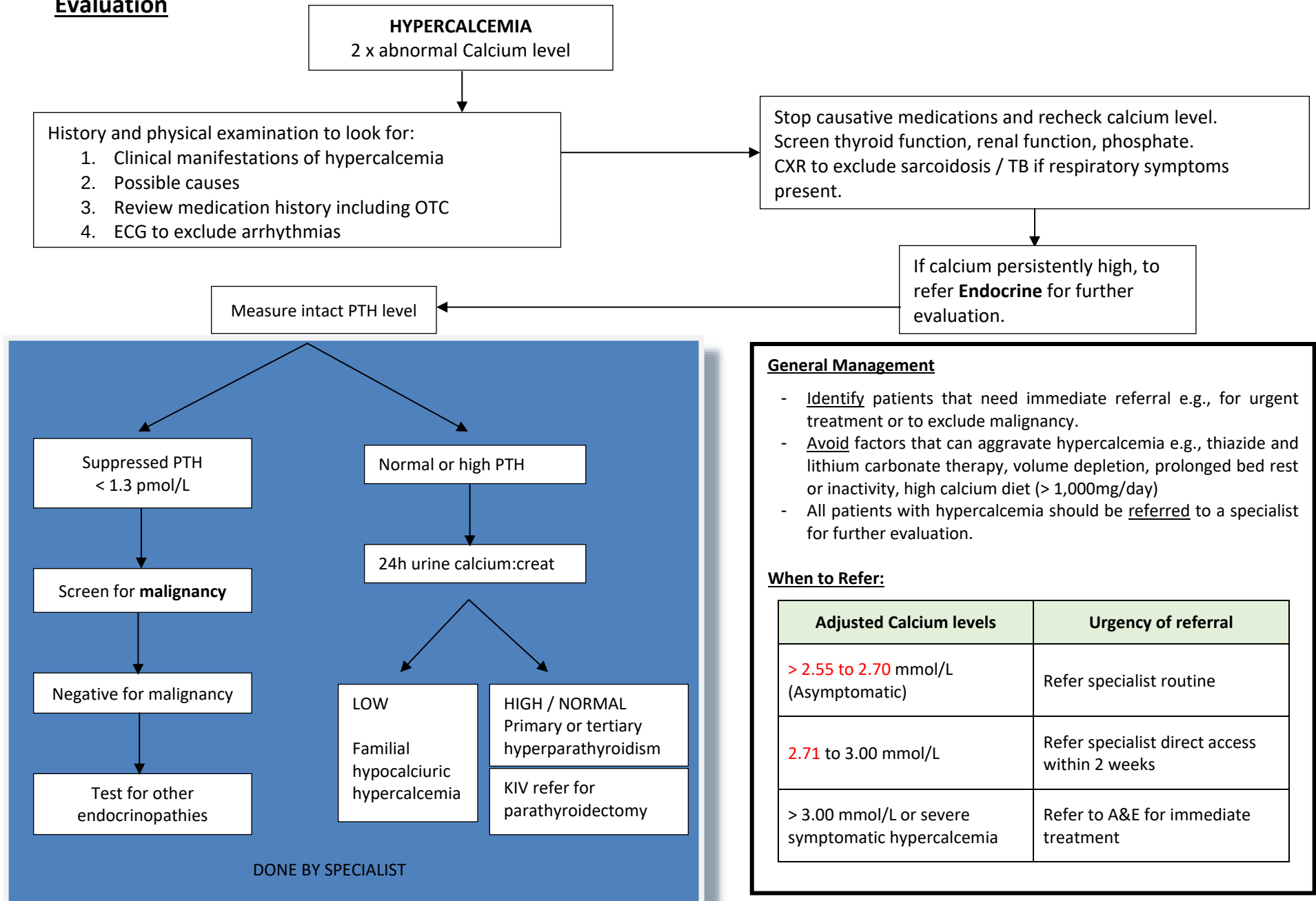
Kidney (Stones)	Bones	Abdominal Groans (Gastrointestinal)	Psychic Moans Neuromuscular	Cardiovascular	Others
Dehydration Nephrolithiasis Nephrocalcinosis Nephrogenic diabetes insipidus	Bone Pain Arthritis Osteoporosis	Nausea, vomiting Constipation Abdominal pain Pancreatitis Peptic Ulcer disease	Impaired concentration / memory Lethargy / fatigue Muscle weakness Confusion / coma	Hypertension Shortened QT Cardiac arrhythmias	Itch Keratitis Conjunctivitis

Causes

Parathyroid Hormone-Mediated	Non-Parathyroid Hormone Mediated
* <u>Primary hyperparathyroidism</u> - Sporadic adenomas, hyperplasia, carcinoma - Inherited (e.g., Multiple Endocrine Neoplasia syndromes, familial hyperparathyroidism, hyperparathyroidism-jaw tumour syndrome)	* <u>Malignancy associated hypercalcemia</u> - <u>Humoral hypercalcemia of malignancy</u> (<i>PTHrp or 1,25OH vitamin D</i>) (e.g., squamous cell lung / head / neck, breast, ovary, renal cell, bladder, T-cell lymphoma) (benign or malignant pheochromocytomas, islet cell tumours, carcinoids) - <u>Local osteolytic hypercalcemia</u> (multiple myeloma, breast ca, lymphoma)
Tertiary hyperparathyroidism (renal failure or vitamin D deficiency)	Chronic granulomatous disorders (e.g., sarcoidosis, tuberculosis)
Familial hypocalciuric hypercalcemia	<u>Endocrine disorders:</u> - Hyperthyroidism - Adrenal insufficiency
Lithium induced	<u>Medications:</u> - Vitamin D or A intoxication (due to supplements) - Thiazides (exacerbates underlying pathophysiology) - Milk-alkali syndrome (from calcium antacids)
	<u>Miscellaneous:</u> - Immobilisation with high bone turnover (e.g., Paget's disease) - Acute renal failure - Parenteral nutrition - Recovery phase of rhabdomyolysis

***More than 90% of cases are due to primary hyperparathyroidism and malignancy.**

Evaluation



HYPOCALCAEMIA

Causes, Symptoms and Urgency of Referral

* **Common causes:** post-surgical, autoimmune, Vitamin D deficiency, chronic kidney disease.

Other Causes:

LOW OR INAPPROPRIATELY NORMAL PTH	HIGH PTH	OTHERS
Hypoparathyroidism * Post-surgical (thyroidectomy / parathyroidectomy / radical neck dissection) * Autoimmune (idiopathic or polyglandular syndromes) Genetic (Congenital or familial) Radiation Infiltrative (e.g., hemochromatosis, Wilson's) Functional (Hypomagnesemia can impair PTH secretion and action)	* Vitamin D deficiency or resistance - lack of sun, malnutrition, malabsorption - accelerated loss (anticonvulsants, impaired enterohepatic circulation) - impaired hydroxylation (renal, liver failure) - vitamin D dependent rickets type I and II PTH Resistance (Pseudohypoparathyroidism type 1a, 1b) Others: Loss of calcium from circulation.	Loss of calcium from circulation (PTH levels will also be elevated) ▪ Extravascular deposition - osteoblastic metastases, hungry bone syndrome - acute pancreatitis - tumour lysis syndrome - rhabdomyolysis ▪ Chelation - Infusion of phosphate or citrated blood transfusions - foscarnet Drugs (e.g., high-dose intravenous bisphosphonates, cinacalcet, cisplatin, phenytoin, isoniazid) Critical illness

Symptoms

Neuromuscular irritability (perioral paresthesia, tingling of fingers and toes, carpopedal spasms, cramps, tetany)
Seizures, cognitive impairment, personality disturbances

Signs

Previous neck surgery or radiation to head and neck.
Chvostek and Trousseau signs may be elicited.
ECG: prolonged QT intervals

First line investigations in the polyclinic (for routine referrals):

- Serum calcium (corrected for albumin) • Phosphate
- Sodium / Potassium • Creatinine • Alkaline phosphatase
- 25-hydroxyvitamin D • ECG

When to Refer:

Adjusted Calcium levels	Urgency of referral
2.01 to < 2.10 mmol/L	Routine referral
1.80 to 2.00 mmol/L without symptoms	Refer specialist direct access within 2 weeks
< 1.80 mmol/L or symptomatic < 2.00 mmol/L	Refer to A&E for urgent treatment

VITAMIN D DEFICIENCY

Screening and Replacement

Age	Vitamin D requirements/day	Calcium requirements/day
Infants < 1yr old	400 IU	300 - 400 mg
Children 1-18yr old	600 IU	500 - 1000 mg
Adults 19-50yrs old	600 IU	800 mg
Adults who are > 50yrs old, pregnant / breastfeeding	800 IU	1000 mg
Adults with osteoporosis	1000 IU	1200 mg

Vitamin D cut-offs

for patients with osteoporosis

< 10 µg/L:	Severe deficiency
10-19 µg/L:	Deficiency
20-29 µg/L:	Insufficiency (for osteoporosis)
30-100 µg/L:	Sufficiency
> 100 µg/L:	Possible harmful

Who to screen for Vitamin D deficiency?

1. Patients with osteoporosis
2. Risk of frequent falls
3. Pregnant and breastfeeding women
4. Chronic Renal disease (these patients will need calcitriol)
5. Malabsorption
6. Osteomalacia

Manifestations of Vitamin D deficiency:

1. Bone discomfort or pain (often throbbing) in low back, pelvis, lower extremities
2. Increased risk of falls and impaired physical function
3. Muscle aches
4. Proximal muscle weakness
5. Symmetric low back pain in women

Benefits of Vitamin D supplementation

Fall and fracture prevention in elderly patients mainly in the frail with proven deficiency. Routine supplementation in healthy community-dwelling patients without known deficiency has not been shown to reduce falls. Though observational studies have suggested that the risks of cancer, autoimmune and cardiovascular diseases are higher in vitamin D deficiency, a direct causal relationship has not been established.

REPLACEMENT:

1. Cholecalciferol (D3) oral D-cure 50,000 IU per week for 6-8 weeks for vit D < 25 µg/L.
2. Followed by maintenance dose of cholecalciferol (D3) oral liquid D-cure 25,000 IU per month or oral tablets 1,000 IU per day
3. Vitamin D levels should be rechecked about 3 months later to ensure adequate replacement (≥ 20 µg/L for general population, ≥ 30 µg/L if osteoporotic)

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